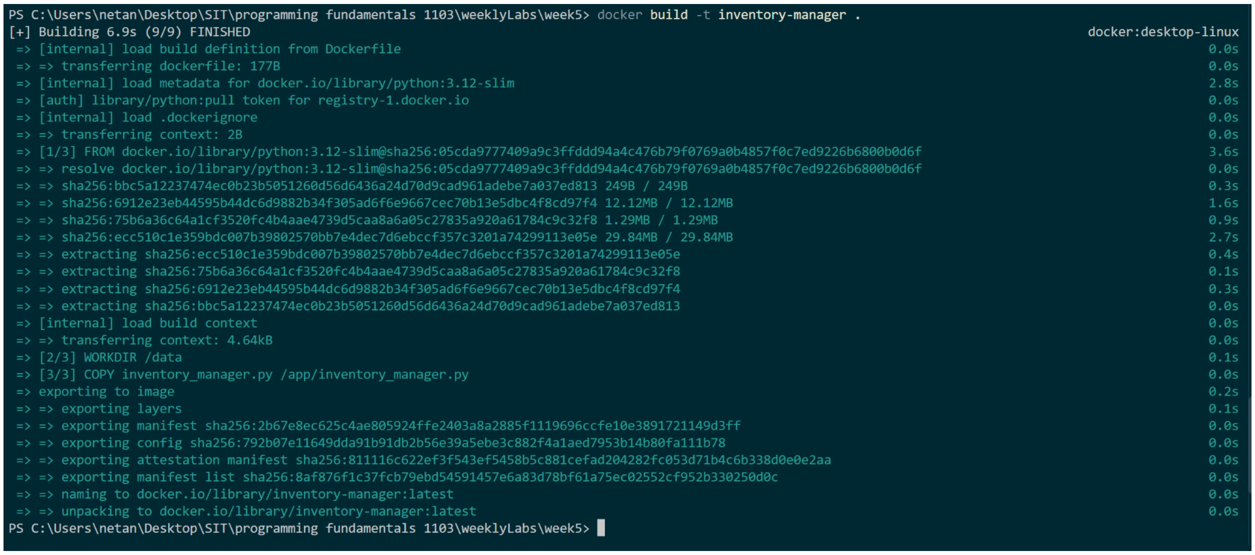
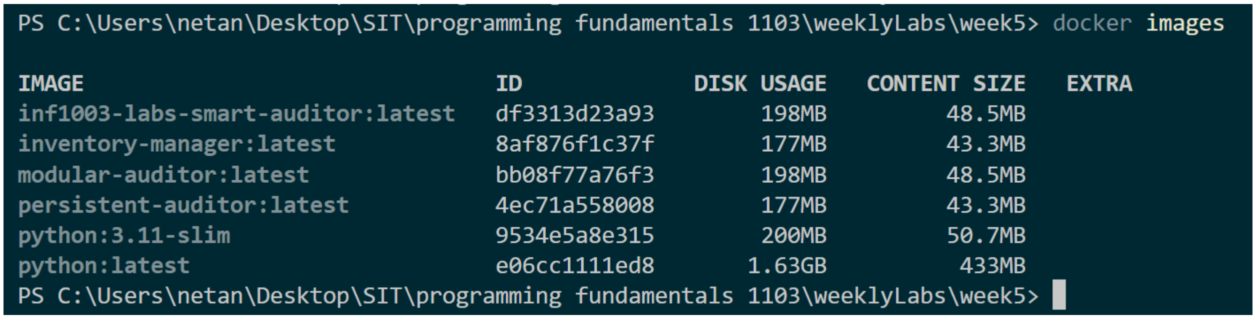
https://github.com/aynaten/inf1103-labs/tree/master

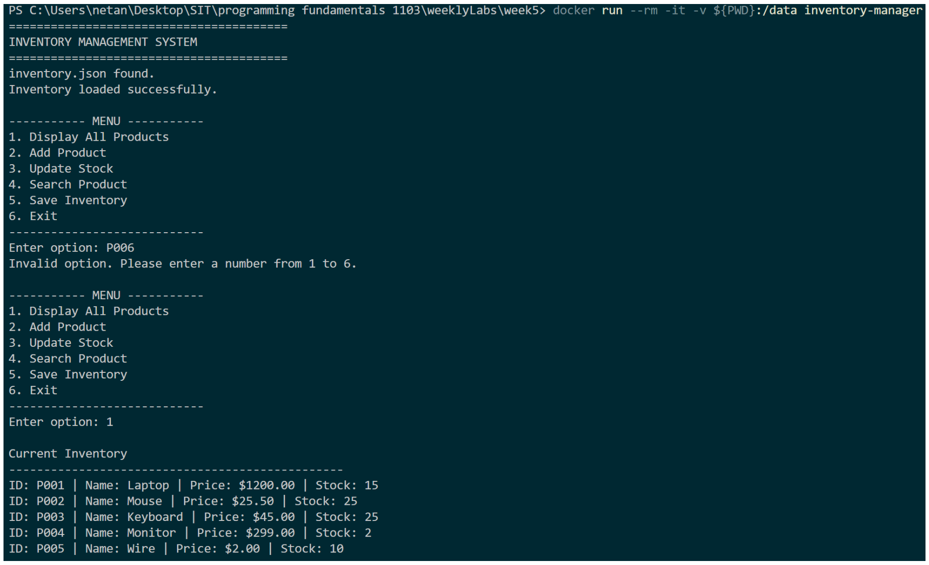


```
PS C:\Users\netan\Desktop\SIT\programming fundamentals 1103\weeklyLabs\week5> docker build -t inventory-manager .
[+] Building 6.9s (9/9) FINISHED                                                                                                               docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                                                           0.0s
 => => transferring dockerfile: 177B                                                                                                                           0.0s
 => [internal] load metadata for docker.io/library/python:3.12-slim                                                                                            2.8s
 => [auth] library/python:pull token for registry-1.docker.io                                                                                                  0.0s
 => [internal] load .dockerignore                                                                                                                              0.0s
 => => transferring context: 2B                                                                                                                                0.0s
 => [1/3] FROM docker.io/library/python:3.12-slim@sha256:05cda9777409a9c3ffddd94a4c476b79f0769a0b4857f0c7ed9226b6800b0d6f                                      3.6s
 => => resolve docker.io/library/python:3.12-slim@sha256:05cda9777409a9c3ffddd94a4c476b79f0769a0b4857f0c7ed9226b6800b0d6f                                      0.0s
 => => sha256:bbc5a12237474ec0b23b5051260d56d6436a24d70d9cad961adebe7a037ed813 249B / 249B                                                                     0.3s
 => => sha256:6912e23eb44595b44dc6d9882b34f305ad6f6e9667cec70b13e5dbc4f8cd97f4 12.12MB / 12.12MB                                                               1.6s
 => => sha256:75b6a36c64a1cf3520fc4b4aae4739d5caa8a6a05c27835a920a61784c9c32f8 1.29MB / 1.29MB                                                                 0.9s
 => => sha256:ecc510c1e359bdc007b39802570bb7e4dec7d6ebccf357c3201a74299113e05e 29.84MB / 29.84MB                                                               2.7s
 => => extracting sha256:ecc510c1e359bdc007b39802570bb7e4dec7d6ebccf357c3201a74299113e05e                                                                      0.4s
 => => extracting sha256:75b6a36c64a1cf3520fc4b4aae4739d5caa8a6a05c27835a920a61784c9c32f8                                                                      0.1s
 => => extracting sha256:6912e23eb44595b44dc6d9882b34f305ad6f6e9667cec70b13e5dbc4f8cd97f4                                                                      0.3s
 => => extracting sha256:bbc5a12237474ec0b23b5051260d56d6436a24d70d9cad961adebe7a037ed813                                                                      0.0s
 => [internal] load build context                                                                                                                              0.0s
 => => transferring context: 4.64kB                                                                                                                            0.0s
 => [2/3] WORKDIR /data                                                                                                                                        0.1s
 => [3/3] COPY inventory_manager.py /app/inventory_manager.py                                                                                                  0.0s
 => exporting to image                                                                                                                                         0.2s
 => => exporting layers                                                                                                                                        0.1s
 => => exporting manifest sha256:2b67e8ec625c4ae805924ffe2403a8a2885f1119696ccfe10e3891721149d3ff                                                              0.0s
 => => exporting config sha256:792b07e11649dda91b91db2b56e39a5ebe3c882f4a1aed7953b14b80fa111b78                                                                0.0s
 => => exporting attestation manifest sha256:811116c622ef3f543ef5458b5c881cefad204282fc053d71b4c6b338d0e0e2aa                                                  0.0s
 => => exporting manifest list sha256:8af876f1c37fcb79ebd54591457e6a83d78bf61a75ec02552cf952b330250d0c                                                         0.0s
 => => naming to docker.io/library/inventory-manager:latest                                                                                                    0.0s
 => => unpacking to docker.io/library/inventory-manager:latest                                                                                                 0.0s
PS C:\Users\netan\Desktop\SIT\programming fundamentals 1103\weeklyLabs\week5>
```



```
PS C:\Users\netan\Desktop\SIT\programming fundamentals 1103\weeklyLabs\week5> docker images

IMAGE                               ID             DISK USAGE   CONTENT SIZE   EXTRA
inf1003-labs-smart-auditor:latest   df3313d23a93        198MB         48.5MB
inventory-manager:latest            8af876f1c37f        177MB         43.3MB
modular-auditor:latest              bb08f77a76f3        198MB         48.5MB
persistent-auditor:latest           4ec71a558008        177MB         43.3MB
python:3.11-slim                    9534e5a8e315        200MB         50.7MB
python:latest                       e06cc1111ed8       1.63GB          433MB
PS C:\Users\netan\Desktop\SIT\programming fundamentals 1103\weeklyLabs\week5>
```

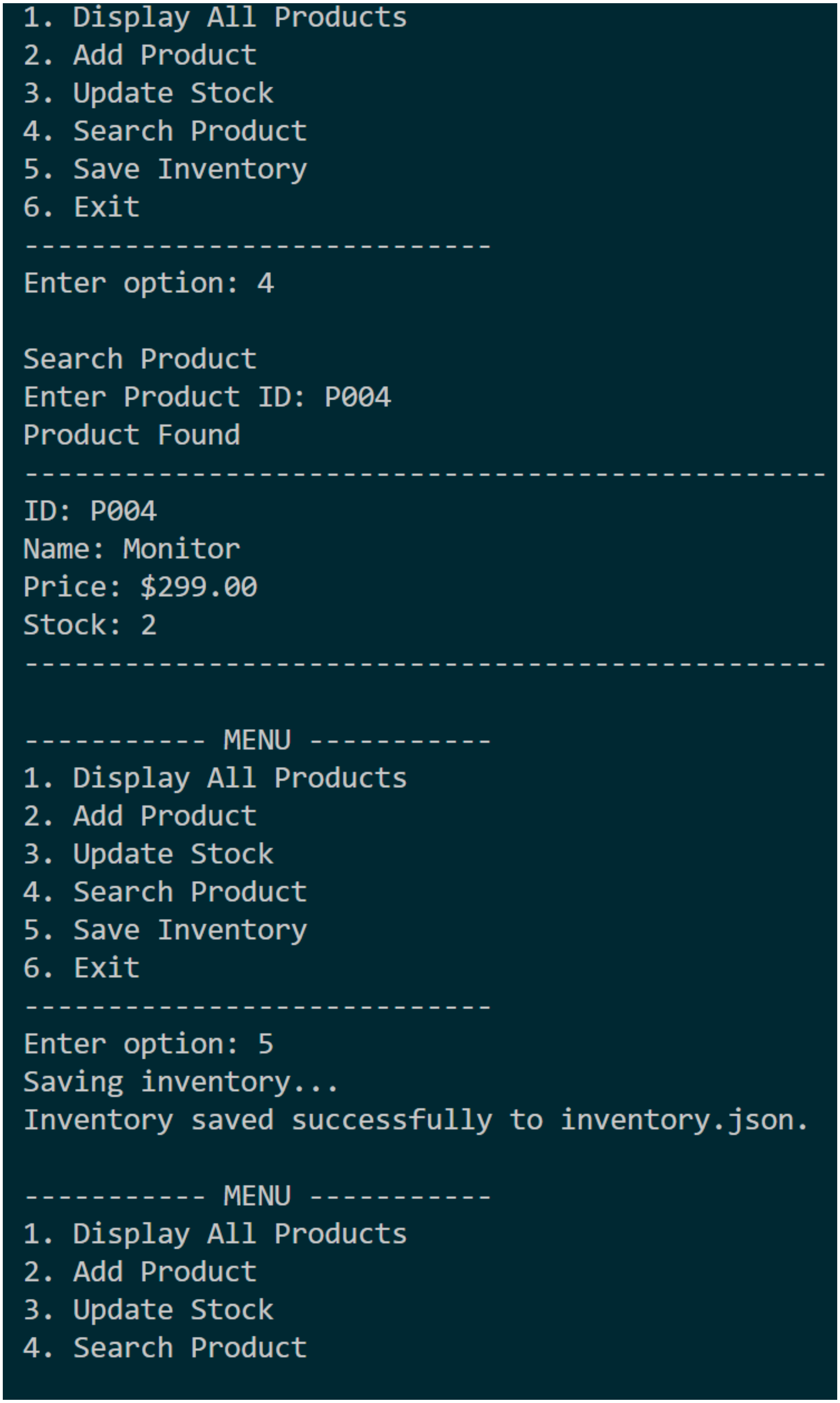


```
PS C:\Users\netan\Desktop\SIT\programming fundamentals 1103\weeklyLabs\week5> docker run --rm -it -v ${PWD}:/data inventory-manager
========================================
INVENTORY MANAGEMENT SYSTEM
========================================
inventory.json found.
Inventory loaded successfully.

----------- MENU -----------
1. Display All Products
2. Add Product
3. Update Stock
4. Search Product
5. Save Inventory
6. Exit
----------------------------
Enter option: P006
Invalid option. Please enter a number from 1 to 6.

----------- MENU -----------
1. Display All Products
2. Add Product
3. Update Stock
4. Search Product
5. Save Inventory
6. Exit
----------------------------
Enter option: 1

Current Inventory
------------------------------------------------
ID: P001 | Name: Laptop | Price: $1200.00 | Stock: 15
ID: P002 | Name: Mouse | Price: $25.50 | Stock: 25
ID: P003 | Name: Keyboard | Price: $45.00 | Stock: 25
ID: P004 | Name: Monitor | Price: $299.00 | Stock: 2
ID: P005 | Name: Wire | Price: $2.00 | Stock: 10
```

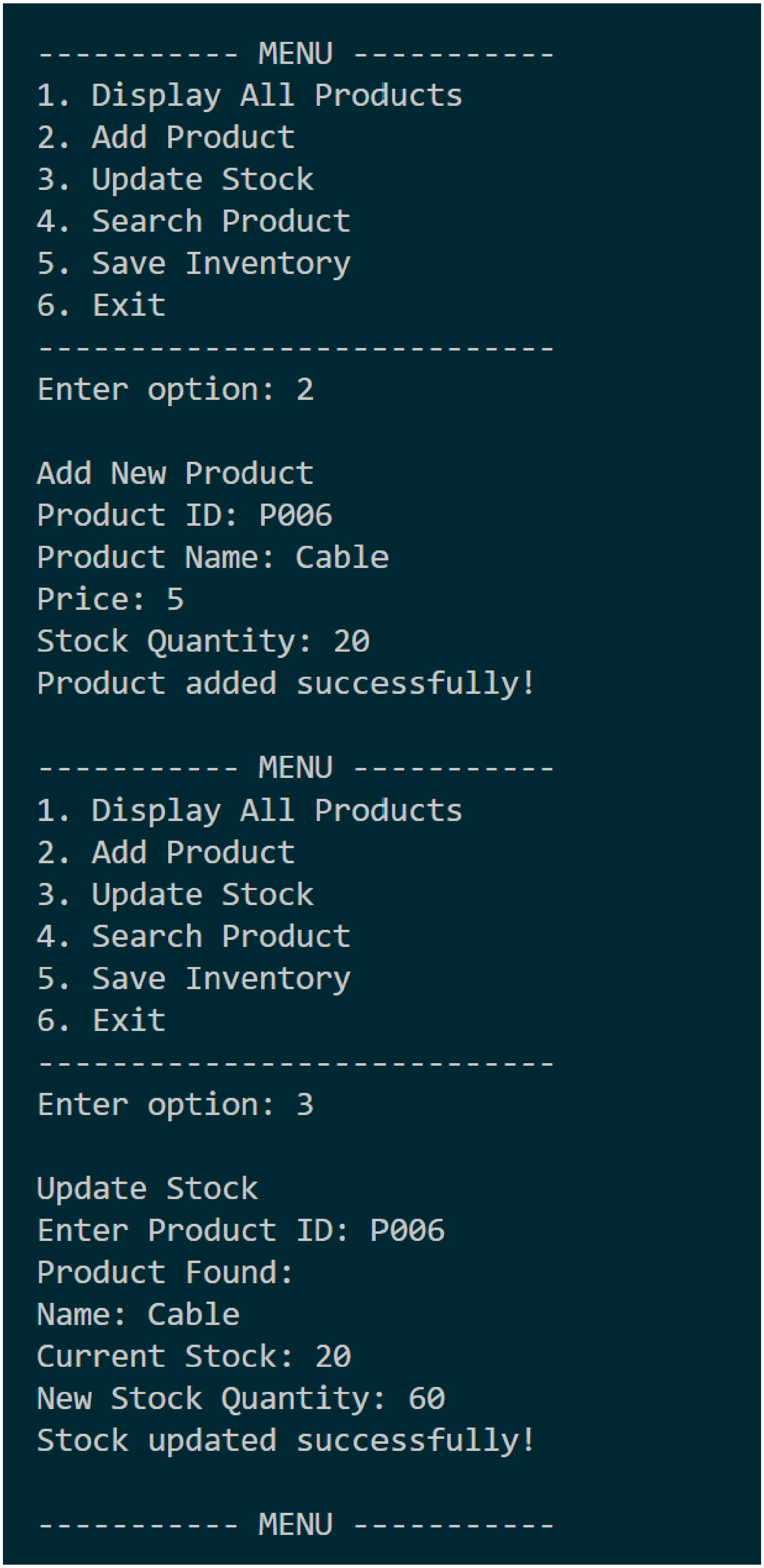


```
1. Display All Products
2. Add Product
3. Update Stock
4. Search Product
5. Save Inventory
6. Exit
----------------------------
Enter option: 4

Search Product
Enter Product ID: P004
Product Found
------------------------------------------------
ID: P004
Name: Monitor
Price: $299.00
Stock: 2
------------------------------------------------

----------- MENU -----------
1. Display All Products
2. Add Product
3. Update Stock
4. Search Product
5. Save Inventory
6. Exit
----------------------------
Enter option: 5
Saving inventory...
Inventory saved successfully to inventory.json.

----------- MENU -----------
1. Display All Products
2. Add Product
3. Update Stock
4. Search Product
```



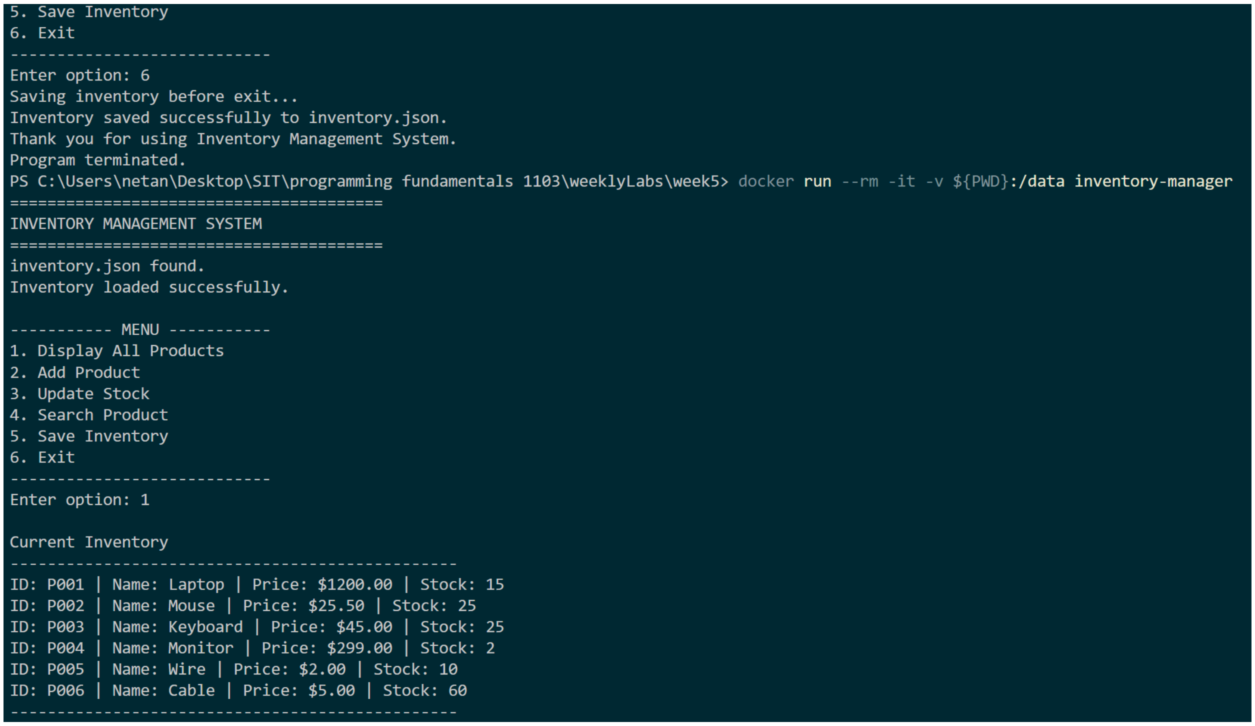
```
----------- MENU -----------
1. Display All Products
2. Add Product
3. Update Stock
4. Search Product
5. Save Inventory
6. Exit
----------------------------
Enter option: 2

Add New Product
Product ID: P006
Product Name: Cable
Price: 5
Stock Quantity: 20
Product added successfully!

----------- MENU -----------
1. Display All Products
2. Add Product
3. Update Stock
4. Search Product
5. Save Inventory
6. Exit
----------------------------
Enter option: 3

Update Stock
Enter Product ID: P006
Product Found:
Name: Cable
Current Stock: 20
New Stock Quantity: 60
Stock updated successfully!

----------- MENU -----------
```



```
5. Save Inventory
6. Exit
----------------------------
Enter option: 6
Saving inventory before exit...
Inventory saved successfully to inventory.json.
Thank you for using Inventory Management System.
Program terminated.
PS C:\Users\netan\Desktop\SIT\programming fundamentals 1103\weeklyLabs\week5> docker run --rm -it -v ${PWD}:/data inventory-manager
========================================
INVENTORY MANAGEMENT SYSTEM
========================================
inventory.json found.
Inventory loaded successfully.

----------- MENU -----------
1. Display All Products
2. Add Product
3. Update Stock
4. Search Product
5. Save Inventory
6. Exit
----------------------------
Enter option: 1

Current Inventory
------------------------------------------------
ID: P001 | Name: Laptop | Price: $1200.00 | Stock: 15
ID: P002 | Name: Mouse | Price: $25.50 | Stock: 25
ID: P003 | Name: Keyboard | Price: $45.00 | Stock: 25
ID: P004 | Name: Monitor | Price: $299.00 | Stock: 2
ID: P005 | Name: Wire | Price: $2.00 | Stock: 10
ID: P006 | Name: Cable | Price: $5.00 | Stock: 60
------------------------------------------------
```

Re-ran to show that the changes were saved and will be shown the next time docker runs again